Hierarchical Sparse Reconstruction Based Multi-feature Saliency for Target Detection in SAR Images

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***Abstract*—The utilization of clutter information is useful for target detection. However, the clutter statistical modeling, used in conventional constant false alarm rate (CFAR) methods, is usually difficult for a real synthetic aperture radar (SAR) image. And though intensity feature is basic for target detection, only using intensity feature is insufficient considering the speckle noise. To address these problems, we propose a novel hierarchical sparse reconstruction based multi-feature saliency method to detect targets in SAR images. The proposed method hierarchically generates saliency maps using sparse reconstruction based on clutter templates on instance feature and structural feature separately. The clutter templates sampled from the pure clutter SAR images contains the clutter information. Though the sparse reconstruction based on clutter templates, we can fully utilize clutter information without clutter statistical modeling. Moreover, the proposed method jointly uses the instance feature and the structure feature, making the saliency detection more robust to speckle noise. We evaluate the proposed method on the miniSAR real data. The experimental results demonstrate that the proposed method outperforms conventional SAR target detection methods in terms of the precision, recall and F1-score.**

Keywords—sparse reconstruction, saliency, synthetic aperture radar (SAR), target detection

I. INTRODUCTION

Because of the all-weather and day-and-night characteristics, synthetic aperture radar (SAR) plays an increasingly crucial role in both military and civilian areas. And the resolution of SAR images becomes higher and higher with the development of SAR sensor technology. Nowadays, the SAR interpretation, known as automatic target recognition (ATR), has become one of the most challenging field with a great deal of SAR images available.

In the SAR ATR system, target detection stage isolates the regions of interest from the large-scene SAR images, which is becoming more and more important. There are many SAR target detection algorithms available in literatures in recent years. The constant false alarm rate (CFAR) methods [1, 2] have been extensively studied for SAR target detection. The two-parameter CFAR detector [1] based on the Gaussian background statistical distribution is very popular. The Gamma-based CFAR detector which assumes the background clutter obeys a Gamma distribution is then proposed by [2]. These conventional SAR target detection methods use clutter information through clutter statistical modeling. Nevertheless, for a real SAR image of which the scene is complex, an appropriate clutter statistical distribution model [3] is difficult to select. And if an inappropriate clutter model is selected, the performance of target detection will decrease.

In recent years, saliency detection attracts considerable attention in many fields, like machine translation and visual question answering. Saliency detection, which aims at highlighting targets while suppressing clutter, has been widely applied in many fields for rapid scene understanding. Many novel saliency detection methods have been proposed recently. Itti *et al.* [4] proposed a biologically plausible model to compute center-surround difference. Modified by the Itti's method, Liu *et al.* [5] integrated the singular value decomposition (SVD) into the Itti's model and achieved better performance in saliency detection. Wang *et al.* [6] regarded the superpixels as units and computed the global spatially dissimilarity of every superpixel to generate a saliency map on superpixel-level. Hou *et al.* [7] based on the hypothesis that saliency is the residual difference between the perceived spectrum and the characteristic spectrum of natural images and then proposed a spectral residual (SR) method. Regarding the difference between targets and background clutter as saliency cue, clutter templates are extracted at first and then the difference of every sample is computed as the salient value [8-14]. For clutter templates, due to the fact that people always place targets in the center of optical images, the boundary prior is proposed that the boundary of optical images is always background clutter [15]. And the more different the sample to be detected and clutter templates are, the more likely the sample is target sample. Sparse reconstruction as a measurement to compute difference between sample to be detected and clutter templates, the sparse reconstruction based saliency detection method [12, 13] is flourishing and achieves great performance in optical images. Based on the boundary prior, many clutter templates are extracted from the boundary of optical images and construct a clutter dictionary. Then, on the basic of the clutter dictionary, the reconstruction residual of a sample is regarded as the measurement of its salient value.

The sparse reconstruction based saliency detection method is expected to introduce into the target detection on SAR images. The clutter information reflected in clutter templates can be fully utilized, at the same time, the clutter statistical modeling can be avoided. Nevertheless, it cannot be directly applied to the SAR images. There are two important reasons. First, there is not a boundary prior in SAR images. The SAR scenes are large-scale and there is no prior information about the locations of targets. Second, only utilizing the intensity information will limit the performance

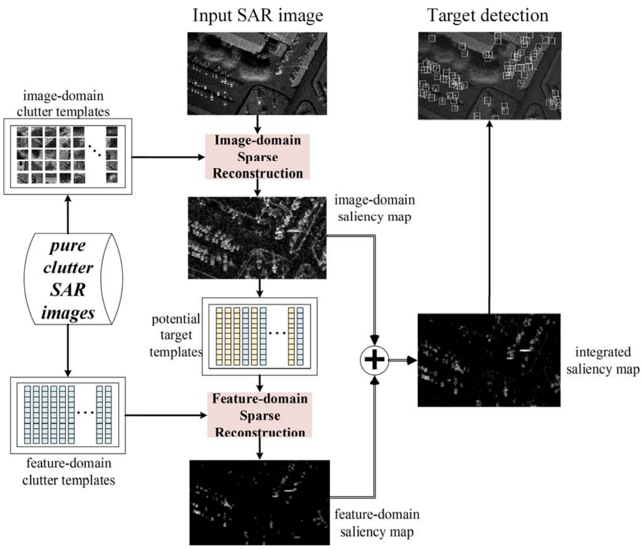

Fig.1. Flowchart of the proposed target detection method in SAR images.

and the intensities of pixels are fluctuant due to the influence of speckle noise.

According to the above-mentioned analyses, in this paper, we propose a hierarchical sparse reconstruction based multi-feature saliency method to detect targets in SAR images. The main contributions are the following.

- The sparse reconstruction based saliency is introduced into the target detection in SAR images. It not only makes full use of clutter information, but also avoids the clutter statistical modeling.
- We jointly utilize the intensity feature and the structure feature, considering the speckle noise in SAR images. We hierarchically generate the image-domain saliency maps using the intensity descriptor and the feature-domain saliency map using the structure information. Through integrating the two saliency map, the proposed target detection method is more robust to speckle noise.
- The clutter templates are obtained from the pure clutter SAR images which is more suitable, and we employ different dictionary constructions at different stages. At the stage of generating image-domain saliency map, we only use clutter templates to construct the dictionary. Since some potential target templates which can provide additional target information can be obtained after the generation of the image-domain saliency map. At the stage of generating feature-domain saliency map, we construct two dictionaries, one is the clutter dictionary with clutter templates, and the other is the mixed dictionary with not only clutter templates but also potential target templates, for reducing false alarms.

The experiments on the miniSAR real data demonstrate that the proposed target detection method achieves the better detection performance than other compared methods. The remainder of this paper is organized as follows. Section II introduces the proposed novel target detection method. Section III gives the experimental results and analysis based

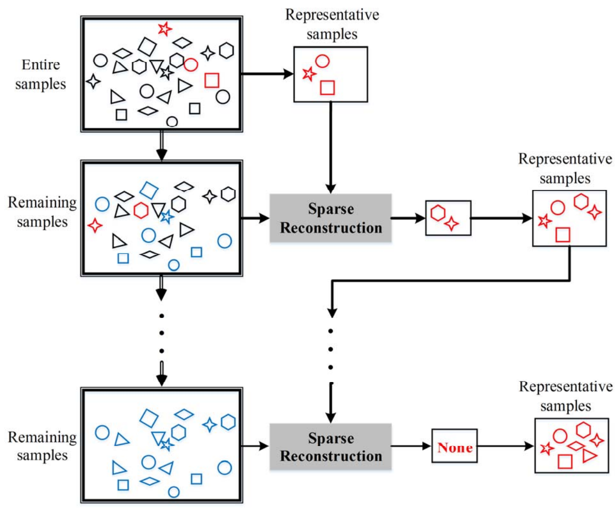

Fig.2. Flowchart of the sample selection method. In the flowchart, the red ones stand for the selected representative samples, the blue ones stand for the samples which can be well sparse reconstructed, and the black ones stand for the samples which cannot be well sparse reconstructed.

on the miniSAR real data. Finally, Section IV concludes this paper.

II. Hierarchical Sparse Reconstruction Based Multi-feature Saliency for Target Detection

Fig. 1 shows the flowchart of the proposed hierarchical sparse reconstruction based multi-feature saliency method on SAR images. The proposed target detection method consists four steps, i.e., the acquirement of clutter templates, the construction of image-domain saliency map, the construction of feature-domain saliency map, the fusion of saliency maps and the location of targets. The four steps are concretely introduced in the following sub-sections.

A. The acquirement of clutter template

The boundary prior, an effective basic rule in optical images, is obviously not suitable for SAR images to extract clutter templates. Therefore, we use the pure clutter SAR images to extract the clutter templates rather than the boundary prior. At first, lots of samples can be sampled from the pure clutter SAR images, and all these samples are clutter samples. Considering that the volume of clutter samples is enormous and there are redundancies, then, we apply the sample selection method to select representative clutter samples from the entire clutter samples [16]. Fig. 2 gives the procedure of the sample selection method. These representative clutter samples can be regarded as the clutter templates.

B. The construction of image-domain saliency map

In the image domain, the pixel-level intensity feature is taken as descriptor, called image-domain descriptor. As usually used in SAR images, the image-domain descriptor of a pixel $\mathbf{x} \in \mathbb{R}^2$ can be represented by a vector of intensity values in the region which is centered on the pixel $\mathbf{x}$. And then, we utilize sparse reconstruction theory to acquire the image-domain saliency map. At First, with the image-domain descriptor, we extract image-domain clutter templates from pure clutter SAR images, which is illuminated in Section II-A, and these image-domain clutter templates are used to construct the image-domain clutter dictionary

$\mathbf{D}^I = \left[\mathbf{d}_1{}^I, \mathbf{d}_2{}^I, \bullet\bullet\bullet, \mathbf{d}_{N_I}{}^I\right] \in \mathbb{R}^{m^2 \times N_I}$, where $\mathbf{d}_j{}^I \in \mathbb{R}^{m^2 \times 1}$, $j \in [1, N_I]$ is an image-domain clutter template, N_I is the number of clutter template. Then, for each pixel $\mathbf{x}_i$, of which image-domain descriptor is represented as $\mathbf{I}(\mathbf{x}_i)$, we encode the $\mathbf{I}(\mathbf{x}_i)$ based on the dictionary $\mathbf{D}^I$. The sparse coding is formulated as

$$\boldsymbol{\alpha}_i^* = \underset{\boldsymbol{\alpha}_i}{\arg\min} \| \mathbf{I}(\mathbf{x}_i) - \mathbf{D}^I \boldsymbol{\alpha}_i \|_2^2 + \lambda \| \boldsymbol{\alpha}_i \|_1 \quad (1)$$

where $\| \bullet \|_1$ denotes the l_1-norm, $\| \bullet \|_2$ denotes the l_2-norm and λ is a regularization parameter. Finally, the sparse reconstruction residual is represented as

$$\varepsilon_i^I = \| \mathbf{I}(\mathbf{x}_i) - \mathbf{D}^I \boldsymbol{\alpha}_i^* \|_2^2 \quad (2)$$

The reconstruction residual ε_i^I reflects how well the reconstruction is performing and it can be regarded as the measurement of salient value. The larger the reconstruction residual is, the more the saliency of the sample to be detected is. By computing the sparse reconstruction residual of every pixel, the image-domain saliency map $\mathbf{Sm}^I$ can be obtained.

C. The construction of feature-domain saliency map

Considering the speckle noise, we utilize not only the intensity feature but also the structure feature. The dense scale invariant feature transform (SIFT) feature which describes the structure information can be a complement to the image-domain descriptor [17]. Therefore, with the superpixels as the elementary units in feature domain, the dense SIFT feature is extracted [16], and then we encode the dense SIFT feature to mid-level feature through the locality-constrained linear coding (LLC) model [18] as feature-domain descriptor. Furthermore, since the potential target information, which is helpful for highlighting targets, can be extracted from the image-domain saliency map, a method which is on the basic of sparse reconstruction and can consider both clutter information and potential target information is proposed. At First, with the LLC descriptor, feature-domain clutter templates $\left[\mathbf{d}_1{}^{FC}, \mathbf{d}_2{}^{FC}, \bullet\bullet\bullet, \mathbf{d}_{N_C}{}^{FC}\right] \in \mathbb{R}^{l \times N_C}$ are extracted from pure clutter SAR images, where $\mathbf{d}_j{}^{FC} \in \mathbb{R}^{l \times 1}, j \in [1, N_C]$ is a feature-domain clutter template, N_C is the number of feature-domain clutter templates, l is the dimension of LLC feature. Secondly, some potential target superpixels which are salient in image-domain saliency map are selected. The LLC features of the potential target superpixels are extracted as the potential target templates $\left[\mathbf{d}_1{}^{FT}, \mathbf{d}_2{}^{FT}, \bullet\bullet\bullet,, \mathbf{d}_{N_S}{}^{FT}\right] \in \mathbb{R}^{l \times N_S}$, where $\mathbf{d}_j{}^{FT} \in \mathbb{R}^{l \times 1}$, $j \in [1, N_S]$ is a feature-domain potential target template, N_S is the number of potential target templates. Thirdly, based on the clutter templates and the potential target templates, we construct two different dictionaries. One of the dictionaries is constructed only with the clutter templates, which is represented as $\mathbf{D}^{FC} = \left[\mathbf{d}_1{}^{FC}, \mathbf{d}_2{}^{FC}, \bullet\bullet\bullet, \mathbf{d}_{N_C}{}^{FC}\right] \in \mathbb{R}^{l \times N_C}$. The other dictionary contains not only the clutter templates but also the potential target templates, which is represented as $\mathbf{D}^{FE} = \left[\mathbf{d}_1{}^{FC}, \mathbf{d}_2{}^{FC}, \bullet\bullet\bullet, \mathbf{d}_{N_C}{}^{FC}, \mathbf{d}_1{}^{FT}, \mathbf{d}_2{}^{FT}, \bullet\bullet\bullet, \mathbf{d}_{N_S}{}^{FT}\right] \in \mathbb{R}^{l \times N_E}$, where $N_E = N_C + N_S$ is the number of templates in dictionary $\mathbf{D}^{FE}$. Then, for every superpixel $\mathbf{s}_i$, $i \in [1, N_S]$, of which structure feature is represented as $\mathbf{f}(\mathbf{s}_i)$ (similar to $\mathbf{d}_i{}^{FT}$), the sparse representations and reconstruction residuals under two dictionaries are respectively formulated as

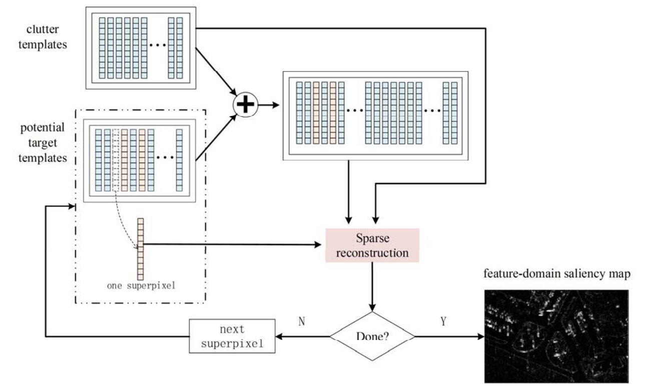

Fig. 3. Flowchart of the generation of feature-domain superpixel-level saliency map.

$$\boldsymbol{\beta}_i^* = \underset{\boldsymbol{\beta}_i}{\arg\min} \| \mathbf{f}(\mathbf{s}_i) - \mathbf{D}^{FC} \boldsymbol{\beta}_i \|_2^2 + \lambda \| \boldsymbol{\beta}_i \|_1 \quad (3)$$

$$\varepsilon_i^{FC} = \| \mathbf{f}(\mathbf{s}_i) - \mathbf{D}^{FC} \boldsymbol{\beta}_i^* \|_2^2 \quad (4)$$

$$\boldsymbol{\gamma}_i^* = \underset{\boldsymbol{\gamma}_i}{\arg\min} \| \mathbf{f}(\mathbf{s}_i) - \mathbf{D}^{FE'} \boldsymbol{\gamma}_i \|_2^2 + \lambda \| \boldsymbol{\gamma}_i \|_1 \quad (5)$$

$$\varepsilon_i^{FE} = \| \mathbf{f}(\mathbf{s}_i) - \mathbf{D}^{FE'} \boldsymbol{\gamma}_i^* \|_2^2 \quad (6)$$

where $\mathbf{D}^{FE'}$ is the same with $\mathbf{D}^{FE}$ except that $\mathbf{D}^{FE'}$ does not contain the $\mathbf{d}_i{}^{FT}$.

The difference between the two above-mentioned sparse representation residuals is represented as

$$\vartriangle\varepsilon_i^F = \left| \varepsilon_i^{FC} - \varepsilon_i^{FE} \right| \quad (7)$$

The $\vartriangle\varepsilon_i^F$ represents the salient value of the superpixels $\mathbf{s}_i$, $i \in [1, N_S]$ and the salient value of pixel which contains in the superpixel $\mathbf{s}_i$ is set to be equal to that of the superpixel. Therefore we can get the feature-domain saliency map $\mathbf{Sm}^F$.

D. The fusion of saliency maps and the location of targets

For taking the intensity information and the structural information into consideration simultaneously, we integrate the saliency map $\mathbf{Sm}^I$ and the saliency map $\mathbf{Sm}^F$ to obtain the final saliency map. Then the winner-take-all approach is employed on the final saliency map to detect targets.

III. EXPERIMENTAL RESULTS AND ANALYSIS

The real SAR data for experiment are the measured miniSAR FARAD dataset and miniSAR dataset, which are acquired by U.S. Sandia National Laboratories. We select one miniSAR FARAD image and one miniSAR image for our experiments and we refer to them as Image I and Image II respectively. Fig. 4 shows the two original SAR images and their ground truths respectively, in which the vehicles are the targets of interest and the others are regarded as clutter.

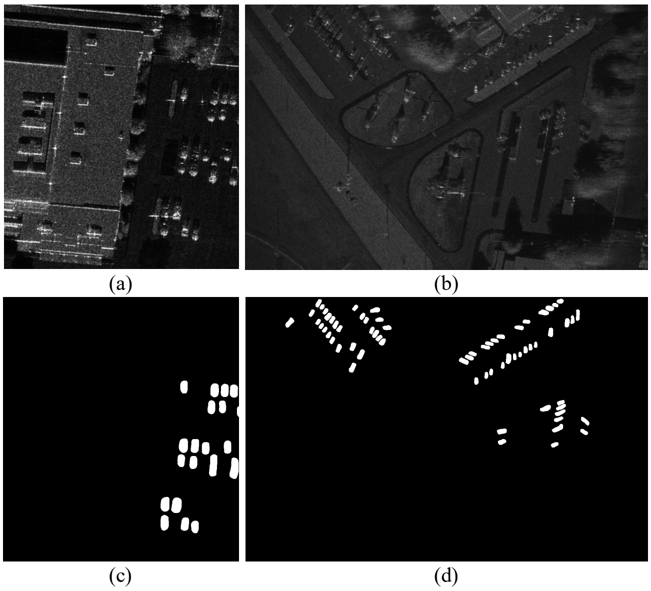

Fig. 4. Original SAR images and their corresponding pixel-level ground truths. (a) is the original miniSAR FARAD image (Image I), (b) is the original miniSAR image (Image II). (c) is the ground truth of the Image I and (d) is the ground truth of theImage II.

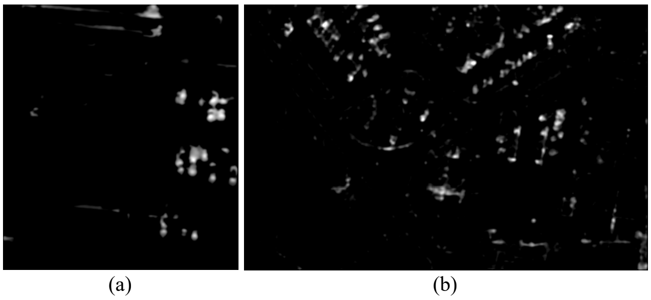

Fig. 5. Saliency maps obtained from the proposed method. (a) is the saliency map on the Image I, (b) is the saliency map on the Image II.

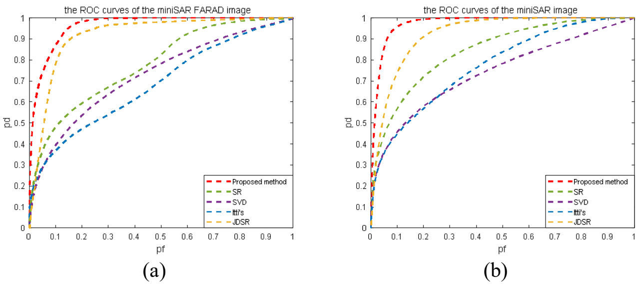

Fig. 6. The ROC curves of different saliency detection methods on the miniSAR FARAD image and the miniSAR image. Here the red line represents the ROC curve of the proposed method.

TABLE I . AUCS OF DIFFERENT SALIENCY DETECTION METHODS

Image	Our Method	SR	SVD	Itti's
Image I	0.9624	0.7786	0.7272	0.6835
Image II	0.9738	0.8450	0.7406	0.7732

A. Evaluation of saliency maps

Visually, the saliency maps in Fig. 5 highlight the vehicle targets and suppress most of clutter for both two SAR images. To quantitatively evaluate the effectiveness of saliency models, Fig. 6 displays the saliency detection performance of the proposed method and other saliency methods, i.e., the SR [7] method, the SVD [5] method and the Itti's [4] method in terms of the Receiver Operating Characteristic curves and we can clearly see that the proposed method, represented by red line, has the highest detection probability in a given false alarms probability on both Image I and Image II. Table I lists their corresponding values of Area Under ROC Curve (AUC). We can observe that the values of AUC of the proposed method are highest, which increases by at least 19.4% on Image I and 12.1% on Image II respectively.

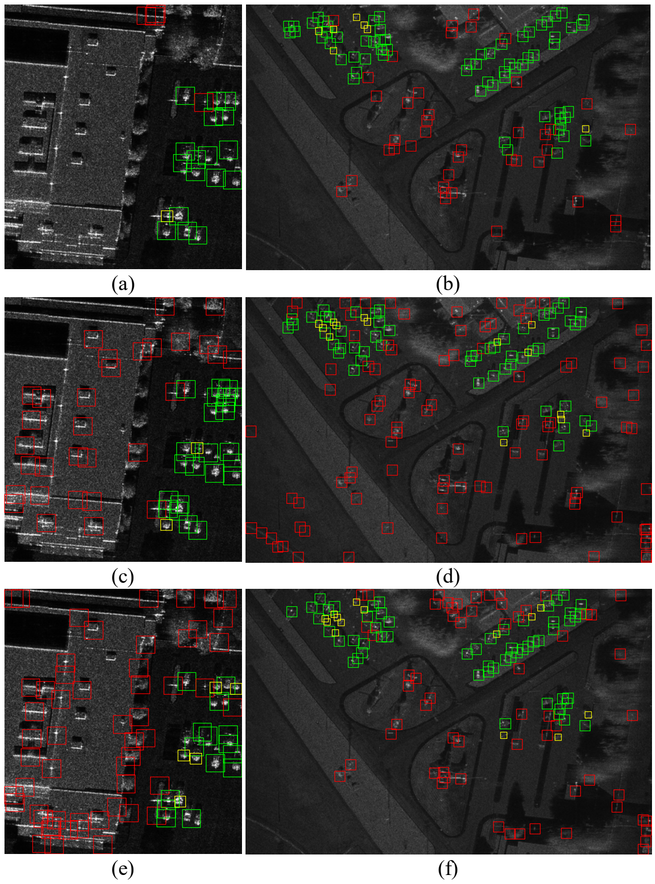

Fig.7. The target detection results on the Image I and the Image II, where green rectangles represent the detected target chips, red rectangles represent false alarms and yellow rectangles represent the missing alarms. (a) and (b) are the proposed method. (c) and (d) are the two parameters CFAR. (e) and (f) are the Gamma-CFAR.

TABLE II.
OVERALL EVALUATION OF DIFFERENT TARGET DETECTION METHODS

	Target amounts	Missing alarms	False alarms	Chip amounts	Recall	Precision	F1-score
Our Method		8	44	128	0.9059	0.6563	0.7611
Gaussian CFAR	85	16	126	203	0.8118	0.3793	0.5170
Gamma CFAR		19	117	189	0.7765	0.3810	0.5112

B. Evaluation of target detection results

In order to validate the effectiveness of the proposed method, conventional SAR target detection methods, including the two parameters CFAR [1] and the Gamma-CFAR [2] are applied to the miniSAR images. Fig. 7 shows the intuitional target detection results of the three methods on the Image I and the Image II. It is found in Fig. 7 that the proposed target detection method improves the overall performance in both two images, which has fewest false alarms and missing alarms.

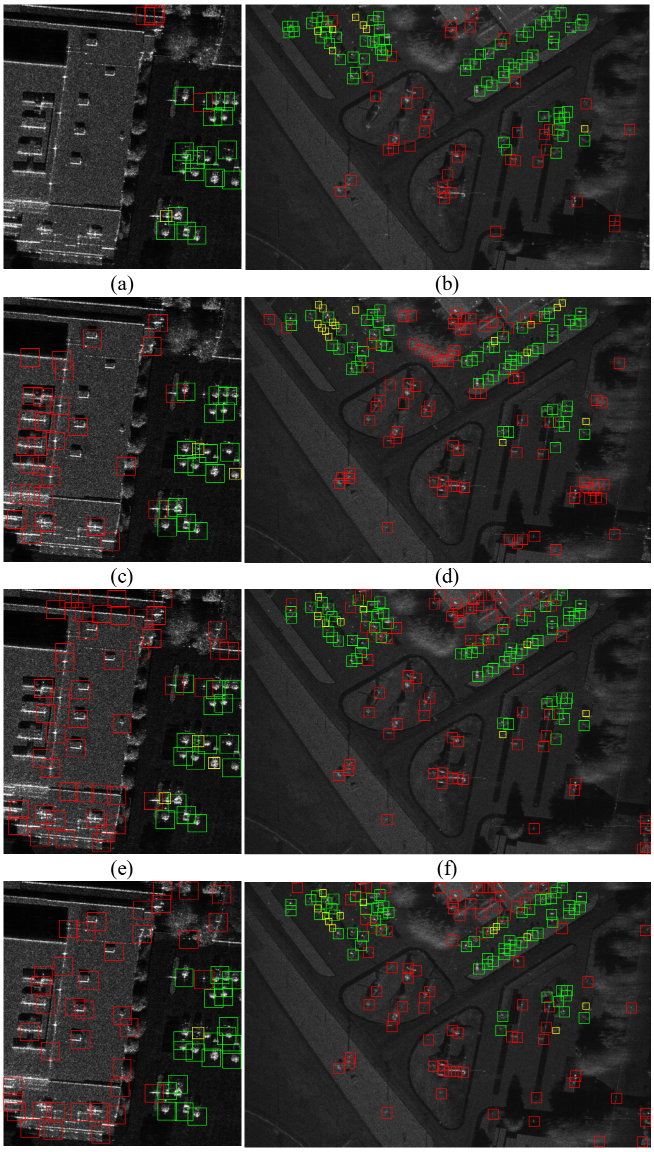

Fig.8. The target detection results on the Image I and the Image II. (a) and (b) are the proposed method. (c) and (d) are the Itti's method. (e) and (f) are the SVD method. (g) and (h) are the SR method.

For quantitative analyses, we evaluate the target detection results in terms of recall, precision and F1-score. The definitions of those evaluation criteria are as follows

$$precision = \frac{\text{the number of target chips}}{\text{the total number of chips}} \tag{8}$$

$$recall = \frac{\text{the number of detected targets}}{\text{the total number of targets}} \tag{9}$$

$$F1-score = \frac{2 \times precision \times recall}{precision + recall} \tag{10}$$

Table II records the overall evaluation of the proposed method and the conventional CFAR methods on Image I and Image II together. It is clearly seen from Table II that the proposed method increases by 9% in recall, 27% in precision and 20% in F1-score. That is, the proposed method performs better compared with conventional CFAR methods.

TABLE III.
OVERALL EVALUATION OF DIFFERENT SALIENCY METHODS FOR TARGET DETECTION

	Target amounts	Missing alarms	False alarms	Chip amounts	Recall	Precision	F1-score
Our Method	85	8	44	128	0.9059	0.6563	0.7611
Itti's Method		18	110	181	0.7882	0.3923	0.5239
SVD Method		11	113	193	0.8706	0.4145	0.5616
SR Method		13	115	197	0.8470	0.4162	0.5581

C. Comparison with saliency methods on target detection

In this section, we compare the proposed method with three saliency detection methods, mentioned in Section III-A, on target detection in Fig. 8 and Table III.

Visually, the proposed target detection method in Fig.8 has fewest false alarms and missing alarms qualitatively. And in Table III, the proposed method gains the best detection performance, which has an at least 3% increase in recall, an at least 24% increase in precision and an about 20% increase in F1-score.

IV. CONCLUSION

In this paper, we propose a novel hierarchical sparse reconstruction based multi-feature saliency method to detect targets in SAR images. The proposed method hierarchically generates saliency maps using sparse reconstruction based on clutter templates on instance feature and structural feature separately. The clutter templates sampled from the pure clutter SAR images contains the clutter information. Though the sparse reconstruction based on clutter templates, we can fully utilize clutter information without clutter statistical modeling. Moreover, the proposed method jointly uses the instance feature and the structure feature, making the saliency detection more robust to speckle noise. We evaluate the proposed method on the miniSAR measured data. The experimental results demonstrate that the proposed SAR detection method outperforms conventional SAR target detection and some saliency detection methods for comparison in terms of the precision, recall and F1-score.

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